

## Inventory Management System

### \* Purpose of Inventory Management

The purpose of inventory management is to ensure that the right products are available in the right quantity, at the right time, and at the right cost. It helps businesses control stock efficiently and supports smooth daily operations.

### Main Purposes of Inventory Management

#### 1. Ensure Product Availability

To make sure products are always available when customers need them, avoiding stock shortages.

#### 2. Prevent Overstocking

To avoid keeping too much stock, which increases storage costs and risk of damage or expiration.

#### 3. Reduce Operational Costs

To minimize costs related to purchasing, storage, handling, and waste.

#### 4. Improve Stock Accuracy

To maintain accurate records of stock received, sold, returned, or damaged.

#### 5. Support Better Decision-Making

To provide reports and data that help managers make better purchasing and sales decisions.

#### 6. Improve Customer Satisfaction

To fulfill customer orders quickly and correctly by maintaining proper stock levels.

#### 7. Increase Business Efficiency

To automate stock tracking and reduce manual work, saving time and effort.

#### 8. Prevent Loss and Theft

To monitor stock movement and detect missing or suspicious inventory activity.

#### 9. Support Financial Control

To track inventory value and assist with accounting, budgeting, and profit analysis.

#### 10. Help Business Growth

To provide insights for forecasting demand, planning expansion, and managing multiple branches or warehouses.

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### \* Scope:

-Manage User login & User Permission

-Manage category inventory

-Manage product inventory

-stock movement

-supplier information

-sales transactions for stores or warehouses.

-Report

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- +Core Features:
- Authentication
  - Role Permission
  - Product Management
  - Category Management
  - Inventory/stock management
  - Stock in & stock out tracking
  - Barcode/QR code scanning
  - Supplier management
  - Sales transaction management
  - Invoice & receipt generation
  - Low stock notifications
  - Sales & inventory reports
  - Report

Tech:  
Mobile App (Flutter) + Backend (Python or Laravel API) + MySQL

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#### Inventory Management System Tables & Relationships

- ++ Table list
1. roles
  2. users
  3. customers
  4. suppliers
  5. categories
  6. products
  7. purchases
  8. purchase\_details
  9. sales
  10. sale\_details
  11. stock\_movements
  12. warehouses
  13. product\_transfers

1. roles  
Stores user roles.

| Column      | Type      |
|-------------|-----------|
| id          | bigint    |
| name        | varchar   |
| description | text      |
| disable     | boolean   |
| created_at  | timestamp |
| updated_at  | timestamp |

2. users  
Stores login accounts.

| Column     | Type      |
|------------|-----------|
| id         | bigint    |
| role_id    | bigint    |
| username   | varchar   |
| email      | varchar   |
| password   | varchar   |
| disable    | boolean   |
| created_at | timestamp |
| updated_at | timestamp |

Relationship

- Many users belong to one role
- users.role\_id -> roles.id

### 3. customers

Stores customer information.

| Column     | Type      |
|------------|-----------|
| id         | bigint    |
| user_id    | bigint    |
| name       | varchar   |
| photo      | varchar   |
| phone      | varchar   |
| address    | text      |
| created_at | timestamp |
| updated_at | timestamp |

Relationship

One customer belongs to one user

customers.user\_id -> users.id

### 4. suppliers

Stores supplier information.

| Column     | Type      |
|------------|-----------|
| id         | bigint    |
| name       | varchar   |
| phone      | varchar   |
| email      | varchar   |
| address    | text      |
| created_at | timestamp |
| updated_at | timestamp |

### 5. categories

Stores product categories.

| Column      | Type      |
|-------------|-----------|
| id          | bigint    |
| name        | varchar   |
| description | text      |
| created_at  | timestamp |
| updated_at  | timestamp |

### 6. products

Stores product details.

| Column      | Type      |
|-------------|-----------|
| id          | bigint    |
| category_id | bigint    |
| supplier_id | bigint    |
| name        | varchar   |
| barcode     | varchar   |
| photo       | varchar   |
| cost_price  | decimal   |
| sale_price  | decimal   |
| qty         | integer   |
| unit        | varchar   |
| description | text      |
| created_at  | timestamp |
| updated_at  | timestamp |

#### Relationship

- Many products belong to one category
- Many products belong to one supplier

products.category\_id -> categories.id

products.supplier\_id -> suppliers.id

#### Purchase Module

##### 7. purchases

Stores purchase invoices.

| Column        | Type      |
|---------------|-----------|
| id            | bigint    |
| supplier_id   | bigint    |
| user_id       | bigint    |
| invoice_no    | varchar   |
| purchase_date | date      |
| total_amount  | decimal   |
| status        | varchar   |
| created_at    | timestamp |
| updated_at    | timestamp |

#### Relationship

purchases.supplier\_id -> suppliers.id

purchases.user\_id -> users.id

##### 8. purchase\_details

Stores purchased products.

| Column      | Type    |
|-------------|---------|
| id          | bigint  |
| purchase_id | bigint  |
| product_id  | bigint  |
| qty         | integer |
| cost_price  | decimal |
| subtotal    | decimal |

#### Relationship

purchase\_details.purchase\_id -> purchases.id  
purchase\_details.product\_id -> products.id

## Sales Module

### 9. sales

Stores sales invoices.

| Column         | Type      |
|----------------|-----------|
| id             | bigint    |
| customer_id    | bigint    |
| user_id        | bigint    |
| invoice_no     | varchar   |
| sale_date      | date      |
| total_amount   | decimal   |
| payment_method | varchar   |
| created_at     | timestamp |
| updated_at     | timestamp |

Relationship

sales.customer\_id -> customers.id  
sales.user\_id -> users.id

### 10. sale\_details

Stores sold products.

| Column     | Type    |
|------------|---------|
| id         | bigint  |
| sale_id    | bigint  |
| product_id | bigint  |
| qty        | integer |
| sale_price | decimal |
| subtotal   | decimal |

Relationship

sale\_details.sale\_id -> sales.id  
sale\_details.product\_id -> products.id

## Inventory Module

### 11. stock\_movements

Tracks stock in/out history.

| Column       | Type                    |
|--------------|-------------------------|
| id           | bigint                  |
| product_id   | bigint                  |
| type         | enum(in,out,adjustment) |
| qty          | integer                 |
| reference_id | bigint                  |
| note         | text                    |
| created_at   | timestamp               |

Relationship

stock\_movements.product\_id -> products.id

# Optional Tables

12. warehouses

| Column   | Type    |
|----------|---------|
| id       | bigint  |
| name     | varchar |
| location | text    |

13. product\_transfers

| Column            | Type   |
|-------------------|--------|
| id                | bigint |
| from_warehouse_id | bigint |
| to_warehouse_id   | bigint |
| transfer_date     | date   |

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## Main Relationships Diagram

